

Excel Module Test

I. Scenario & Context

You have recently been hired as a Junior Data Analyst at **CineData Analytics**, a premier consulting group advising major streaming networks and film distribution houses. Our client, a media network launching a new cinematic curation channel, wants to understand the historical landscape of movies to optimize their broadcasting schedule.

II. Data Description

The dataset provided (Module Test dataset.csv) contains historical data across several core attributes:

- **index:** Global row pointer identifier.
- **Year:** The calendar year the movie was officially released.
- **Length:** Total run time duration of the movie in minutes.
- **Title:** The official release title of the film.
- **Subject:** Major genre classification (e.g., Comedy, Horror, Drama).
- **Actor / Actress:** Registered principal male and female leads.
- **Director:** The primary creative director helming the project.
- **Popularity:** Audience score tracking rating metrics (scaled value).
- **Awards:** Binary structural tracking string indicating critical acclaim recognition ('Yes' or 'No').

III. The Assignment

Your submission must be a single, standalone Excel workbook consisting of a sanitized "Data Studio" sheet and a highly dynamic, scannable "Executive Dashboard" sheet.

Part A: Data Transformation (Mandatory)

Before initializing visual rendering paths, you must execute precise data cleaning protocols to avoid architectural skew:

1. **Imputation**
2. **Calculated Bins:** Create a calculated data matrix column termed **"Era Bracket"** leveraging nested IF or LOOKUP operations to isolate timelines:
 - *Classic Era:* Years < 1959

- *Golden Era*: 1960 to 1979
- *Modern Era*: 1980 to 1999
- *Contemporary*: > 2000

Part B: Executive KPI Cards

Isolate **four (4) dynamic summary metrics** absolute-anchored across the top layer of the visual matrix to instantly frame operational data profiles:

Part C: Advanced Analytical Deep Dive (Choose Any 2)

1. **The Critical Pacing Disconnect:** Does long-form cinematography attract awards, or do compressed running speeds maximize popularity indexes
2. **Genre Evolution & Trend Shift:** Are legacy content structures losing commercial traction?
3. **Era-Over-Era Pacing Volatility:** Entertainment tracking teams hypothesize that digital attention limits have systematically shortened feature runtime lengths over chronological eras. Is this true justify?
4. **Director Profiling and Solo vs. Collaborative Filmography:** Production studios often want to identify if a director works best with a single recurring star or if they maintain a highly diverse casting portfolio.

IV. Presentation & Dashboard Standards

- Operational Slicers
- Aesthetic Discipline
- Dashboard Executive Summary Box